iii) Elasticity

The ability of a material to deform under load and return to its original shape when the load is removed.

iv) Plasticity

The ability of a material to deform under load and retain its new shape when the load is removed.

v) Ductility

The ability of a material to be deformed plastically without rupture under tensile load. Materials possesing ductility can be drawn into fine wire.

vi) Brittleness

It is the property of sudden fracture without any visible permanent deformation.

vii) Machinability

The ease with which a given material may be worked or shaped with a cutting tool is called machinability.

viii) Hardness

The ability of material to resist scratching or indentation by another hard body.

ix) Toughness

The ability of material to absorb energy up to fracture during the plastic deformation.

x) Malleability

The ability of material to be deformed plastically without rupture under compressive load. Malleable metals can be hammered and rolled into thin sheets.

xi) Creep

The slow and progressive deformation of a material for long time with a constant stress.

xii) Fatigue

Failure of materials under repeated or reversal stress is called Fatigue.

xiii) Resilience

The capability of a strained body to recover its side and shape, after deformation caused. Especially by compressive stress.

9.5 CONDUCTING MATERIALS

Conductivity is the property of a material by which it allows the flow of electric current. Conducting materials can be classified into low resistivity, high resistivity and zero resistivity.

9.5.1 Conducting Materials and Its Uses

Silver – Used in contact surface of switch gears and circuit breaker points

Copper – winding in electrical wires, and transformer, in wires, hard drawn copper in transmission lines.

Aluminium – flexible wires, bus bars, domestic wiring.

Tungsten – to make filament in bulbs.

Platinum – thermo couple and constant material

Manganese – standard resistances and shunts.

Constantine – thermo couples, rheostats and starters for electric motors.

Nichrome (Ni,ch) – Heating element in Iron box, heater.

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